

Sample End Semester Question Paper - 2025

Subject: Fluid Mechanics | Code: ME303 | Semester: 3

Branch: Mechanical | Regulation: R20 (sample) | Time: 3 Hours | Max M

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Define Reynolds number.
2. What is head loss?
3. Define boundary layer.
4. What is an ideal fluid?
5. State the continuity equation.
6. Define viscosity.
7. State Pascal's law.
8. What is a streamline?
9. State Bernoulli's equation.
10. What is a venturimeter?

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain fluid properties: viscosity, surface tension, compressibility and vapour pressure.
2. Explain Pascal's law. Solve a hydraulic press problem.
3. State and apply Bernoulli's equation with assumptions.
4. Explain flow measurement: venturimeter, orifice meter and pitot tube.
5. Explain laminar and turbulent flow, Reynolds number and the Darcy-Weisbach equation.
6. Explain the boundary layer concept and its applications.